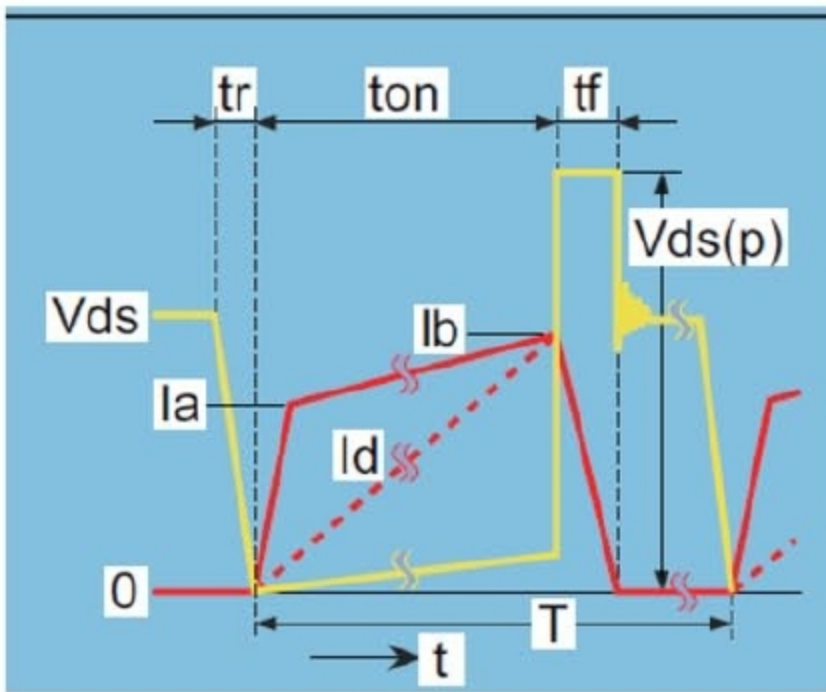


The advantages analysis of Super Junction MOSFET on the high switching frequency LED Driver

In the power electronics field, the big trend in power supply is shrinking the size and weight of magnetic components via increasing the working frequency. The traditional VD-MOSFET (Planar MOSFET) would face problem like high temperature-rising in high frequency application environment. However SJ-MOSFET(Super Junction MOSFET) has a lower C_j (Junction Capacitance), which is more suitable for high switching frequency. Based on Power consumption analysis and practical case, this paper will analyze the application advantages of SJ-MOSFET.

1.MOSFET switching power loss analysis

When Power MOSFET works in the high frequency, the charging and discharging of high input capacitance will lead to large switching loss. The on&off waveform as below:



During T_r , P_t can be ignored comparing with P_{ton} , P_{tf} ;

During Ton:

$$(\text{Full line}) \text{CCM}: P_{ton} = \frac{1}{3} (I_a^2 + I_a \cdot I_b + I_b^2) R_{on} \alpha \cdot t_{on} / T \quad (\text{formula 1})$$

$$(\text{Imaginary line}) \text{DCM}: P_{ton} = \frac{1}{3} \cdot I_b^2 \cdot R_{on} \alpha \cdot t_{on} / T \quad (\text{formula 2})$$

$$\text{During Toff: } P_{toff} = \frac{1}{2} \cdot V_{ds(p)} \cdot I_b \cdot \frac{t_f}{T} \quad (\text{formula 3})$$

α is the $R_{ds(on)}$ temperature coefficient: $\alpha = T(x^\circ\text{C}) / T(25^\circ\text{C})$;

$$T_f = R_g \cdot [C_{gs} + C_{gd}(V_{ds})] \ln \left(\frac{V_{GP}}{V_{TH}} \right) \quad (\text{formula 4})$$

Referring to formula 4, T_f is positive proportion to R_g, C_{gs}, C_{gd} ;

With same $R_{ds(on)}$, the C_{gs}, C_{gd} of SJ-MOSFET will be smaller 2-3 times, and the corresponding turn-off loss is lower 2-3 times than VD-MOSFET.

From formula 3, the higher switching frequency f is, the higher P_{toff} is, if decreasing I_b and t_f , the loss increasing aroused by the frequency increasing can be offset.

Referring to above analysis, SJ-MOSFET is more suitable for higher switching frequency, such as increasing the switching frequency from 65KHZ to 100KHZ-200KHZ.

2.The advantages analysis of high frequency application

a. High frequency Power supply Primary side, MOSFET has a lower current stress

The I_{peak} of classic power supply IC controller UC3842 CCM module is calculated as below:

$$I_{PK_MOSFET} = \frac{P_{IN}}{V_{BULK(min)} \times \frac{N_{PS} \times V_{OUT}}{V_{BULK(min)} + (N_{PS} \times V_{OUT})}} + \left(\frac{V_{BULK(min)}}{2 \times L_m} \times \frac{N_{PS} \times V_{OUT}}{V_{BULK(min)} + (N_{PS} \times V_{OUT})} \right) \cdot \frac{1}{f_{sw}} \quad (\text{formula 5})$$